Vaibhav Pattanshetti

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Education

Pimpri Chinchwad College of Engineering, Pune **Aug 2023 – May 2027 (Expected)**
Bachelor of Technology in Computer Engineering *CGPA: 7.37 / 10*

Technical Skills

Languages & Databases: Python, C, C++, SQL (MySQL)
AI/ML: Machine Learning, Deep Learning, CNNs, Transfer Learning, NLP, LLMs
Libraries: TensorFlow, Keras, scikit-learn, Pandas, NumPy, Matplotlib
GenAI & Agents: LangChain, LangGraph, RAG, MCP (Model Context Protocol), n8n, Ollama
Explainable AI: Grad-CAM, LIME
Tools & Web: Git, GitHub, Docker, Streamlit, Jupyter Notebook, HTML, CSS
Core CS: Object Oriented Programming, Data Structures & Algorithms

Projects

XAI for Multiple Neurological Disorder Identification | *Python, Deep Learning* **GitHub**

- Classified 8 neurological conditions from 24,000+ brain MRI scans using ResNet50 transfer learning with two-phase fine-tuning, data augmentation, and categorical focal loss.
- Addressed class imbalance (503 to 8,104 images per class) using focal loss and computed class weights, achieving 91.99% validation accuracy and 0.93 macro recall, with 0.97–0.99 F1 on all three tumor classes.
- Added Grad-CAM and LIME explainability and built a Streamlit web app for real-time prediction, confidence scores, and visual explanations from uploaded MRI scans.

Agentic RAG Chatbot with Long-Term Memory | *Python, LangGraph* **GitHub**

- Built a multi-tool LangGraph agent with persistent short-term (AsyncSqliteSaver) and long-term memory (InMemoryStore), integrated 5 tools – RAG (FAISS), web search, stock price lookup, calculator and custom MCP expense-tracker server via LangChain's MultiServerMCPClient for automated expense workflows.
- Deployed via a Streamlit UI with real-time token streaming, multi-thread chat history, and live tool-execution status.

Employee Designation & Salary Prediction | *Python, Machine Learning* **GitHub**

- Processed 8300+ employee records (IQR outlier removal, imputation, Ordinal/One-Hot Encoding); used SelectKBest for feature selection, trained 5 regression models and selected LassoCV ($R^2 = 0.90$) for salary prediction.
- Compared 4 classifiers for designation prediction, selected SVM (71.3% accuracy); performed K-Means and Hierarchical clustering to segment salary bands, then deployed via Streamlit and Docker on Docker Hub.

Certifications

- **Database Management Systems** — NPTEL (IIT Kharagpur), *Jan–Mar 2025*: **Top 5% (Elite)** nationwide; relational databases, normalization, SQL, indexing, transactions.
- **Machine Learning using Python** — Simplilearn: Supervised/unsupervised learning, model evaluation, feature engineering on real-world datasets.
- **Power BI for Beginners** — Simplilearn: Interactive dashboards using DAX, Power Query, and data modeling.

Achievements

- **LeetCode**: Solved **200+ DSA problems** (mainly C++); earned the **50 Days Badge 2026**. Profile
- **CodeChef**: Solved **250+ algorithmic problems** across **13 rated contests**; highest rating **1404**. Profile
- **HackerRank**: **5⋆ rating** in **C++, Python, and Problem Solving**. Profile
- **AlphaByte 3.0 Hackathon**: Selected for the final round.